$\phi \in \mathbb{Z}_3^{\mathbb{Z}_3}$ then $3 \cdot \phi(a) = 0$ for all $a \in \mathbb{Z}_3$. Because this group is abelian, we see that it is isomorphic to $\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3$.

c. Let $g_i(x) = (1/c)f_i(x)$ for $i = 1, \cdots, n$. Note that $g_i(a_j) = 0$ for $j \neq i$. Let $d_i = g_i(a_i)$. Note that d_i appears as a product of nonzero factors $(a_i - a_k)$ for $k \neq i$, and consequently $d_i \neq 0$ for $i = 1, \cdots, n$. Let $\phi \in F^F$ and suppose that $\phi(a_i) = c_i$. Let $f(x) \in F$ be defined by

$$f(x) = \sum_{i=1}^{n} \frac{c_i}{d_i} g_i(x).$$

Because $g_j(a_i) = 0$ if $j \neq i$, we see that only the ith term in the sum defining $f(x)$ contributes a nonzero summand to $f(a_i)$. Because $d_i = g_i(a_i)$, we obtain

$$f(a_i) = \frac{c_i}{d_i} g_i(a_i) = \frac{c_i}{d_i} d_i = c_i \text{ for } i = 1, \cdots, n.$$

We have shown that each function ϕ in F^F is a polynomial function $f(x)$ in P_F.

23. Factorization of Polynomials over a Field

1. We perform the desired division.

$$\begin{array}{r|l}
 & x^4 + x^3 + x^2 + x - 2 = q(x) \\
\hline
x^2 + 2x - 3 & x^6 + 3x^5 + \qquad\qquad 4x^2 - 3x + 2 \\
 & \underline{x^6 + 2x^5 - 3x^4} \\
 & \quad x^5 + 3x^4 \\
 & \quad \underline{x^5 + 2x^4 - 3x^3} \\
 & \qquad x^4 + 3x^3 + 4x^2 \\
 & \qquad \underline{x^4 + 2x^3 - 3x^2} \\
 & \qquad\quad x^3 \qquad - 3x \\
 & \qquad\quad \underline{x^3 + 2x^2 - 3x} \\
 & \qquad\qquad - 2x^2 \qquad + 2 \\
 & \qquad\qquad \underline{-2x^2 - 4x + 6} \\
 & \qquad\qquad\qquad 4x + 3 = r(x)
\end{array}$$

2. We perform the desired division.

$$\begin{array}{r|l}
 & 5x^4 + 5x^2 - x = q(x) \\
\hline
3x^2 + 2x - 3 & x^6 + 3x^5 + \qquad\qquad 4x^2 - 3x + 2 \\
 & \underline{x^6 + 3x^5 + 6x^4} \\
 & \qquad\quad x^4 \qquad + 4x^2 \\
 & \qquad\quad \underline{x^4 + 3x^3 + 6x^2} \\
 & \qquad\qquad - 3x^3 - 2x^2 - 3x \\
 & \qquad\qquad \underline{- 3x^3 - 2x^2 + 3x} \\
 & \qquad\qquad\qquad\qquad x + 2 = r(x)
\end{array}$$

3. We perform the desired division.

$$\begin{array}{r} 6x^4+7x^3+2x^2-x+2=q(x) \\ \hline 2x+1 \;\big|\; x^5-2x^4+\qquad\qquad +3x-5 \\ \underline{x^5+6x^4} \\ \quad 3x^4 \\ \quad \underline{3x^4+7x^3} \\ \qquad 4x^3 \\ \qquad \underline{4x^3+2x^2} \\ \qquad\quad -2x^2+3x \\ \qquad\quad \underline{-2x^2-\ x} \\ \qquad\qquad\quad 4x-5 \\ \qquad\qquad\quad \underline{4x+2} \\ \qquad\qquad\qquad\quad 4=r(x) \end{array}$$

4. We perform the desired division.

$$\begin{array}{r} 9x^2+5x+10=q(x) \\ \hline 5x^2-x+2 \;\big|\; x^4+5x^3-3x^2 \\ \underline{x^4+2x^3+7x^2} \\ \quad 3x^3+\ \ x^2 \\ \quad \underline{3x^3-5x^2+10x} \\ \qquad 6x^2+\quad x \\ \qquad \underline{6x^2-10x+9} \\ \qquad\qquad\quad 2=r(x) \end{array}$$

5. Trying $2 \in \mathbb{Z}_5$, we find that $2^2 = 4, 2^3 = 3, 2^4 = 1$, so 2 generates the multiplicative subgroup $\{1, 2, 3, 4\}$ of all units in $\mathbb{Z}^5$. By Corollary 6.16, the only generators are $2^1 = 2$ and $2^3 = 3$.

6. Trying $2 \in \mathbb{Z}_7$, we find that $2^3 = 1$, so 2 does not generate. Trying 3, we find that $3^2 = 2, 3^3 = 6, 3^4 = 4, 3^5 = 5$, and $3^6 = 1$, so 3 generates the six units 1, 2, 3, 4, 5, 6 in $\mathbb{Z}_7$. By Corollary 6.16, the only generators are $3^1 = 3$ and $3^5 = 5$.

7. Trying $2 \in \mathbb{Z}_{17}$, we find that $2^4 = -1$, so $2^8 = 1$ and 2 does not generate. Trying 3, we find that $3^2 = 9, 3^3 = 10, 3^4 = 13, 3^5 = 5, 3^6 = 15, 3^7 = 11, 3^8 = 16 = -1$. Because the order of 3 must divide 16, we see that 3 must be of order 16, so 3 generates the units in $\mathbb{Z}_{17}$. By Corollary 6.16, the only generators are $3^1 = 3, 3^3 = 10, 3^5 = 5, 3^7 = 11, 3^9 = 14, 3^{11} = 7, 3^{13} = 12$, and $3^{15} = 6$.

8. Trying $2 \in \mathbb{Z}_{23}$, we find that $2^2 = 4, 2^3 = 8, 2^4 = 16, 2^5 = 9, 2^6 = 18, 2^7 = 13, 2^8 = 3, 2^9 = 6, 2^{10} = 12$, and $2^{11} = 1$, so 2 does not generate. However, this computation shows that $(-2)^{11} = -1$. Because the order of -2 must divide 22, we see that 21 = -2 must be of order 22, so 21 generates the units of $\mathbb{Z}_{23}$. By Corollary 6.16, the only generators are $(-2)^1 = 21, (-2)^3 = 15, (-2)^5 = 14, (-2)^7 = 10, (-2)^9 = 17, (-2)^{13} = 19, (-2)^{15} = 7, (-2)^{17} = 5, (-2)^{19} = 20$, and $(-2)^{21} = 11$.

9. In $\mathbb{Z}_5$, we have $x^4 + 4 = x^4 - 1 = (x^2 + 1)(x^2 - 1)$. Replacing 1 by - 4 again, we continue and discover that $(x^2 - 4)(x^2 - 1) = (x - 2)(x + 2)(x - 1)(x + 1)$.

10. By inspection, -1 is a zero of $x^3 + 2x^2 + 2x + 1$ in $\mathbb{Z}_7[x]$. Executing the division algorithm as illustrated in our answers to Exercises 1 through 3, we compute $x^3 + 2x^2 + 2x + 1$ divided by $x - (-1) = x + 1$,

and find that

$$x^3 + 2x^2 + 2x + 1 = (x + 1)(x^2 + x + 1).$$

By inspection, 2 and 4 are zeros of $x^2 + x + 1$. Thus the factorization is

$$x^3 + 2x^2 + 2x + 1 = (x + 1)(x - 4)(x - 2).$$

11. By inspection, 3 is a zero of $2x^3 + 3x^2 - 7x - 5$ in $\mathbb{Z}_{11}[x]$. Dividing by $x - 3$ using the technique illustrated in our answers to Exercises 1 through 3, we find that

$$2x^3 + 3x^2 - 7x - 5 = (x - 3)(2)(x^2 - x - 1).$$

By inspection, -3 and 4 are zeros of $x^2 - x - 1$, so the factorization is

$$2x^3 + 3x^2 - 7x - 5 = (x - 3)(x + 3)(2x - 8).$$

12. By inspection, -1 is a zero of $x^3 + 2x + 3$ in $\mathbb{Z}_5[x]$, so the polynomial is not irreducible. We divide by $x + 1$, using the technique of Exercises 1 through 3, and obtain

$$x^3 + 2x + 3 = (x + 1)(x^2 - x + 3).$$

By inspection, -1 and 2 are zeros of $x^2 - x + 3$, so the factorization is

$$x^3 + 2x + 3 = (x + 1)(x + 1)(x - 2).$$

13. Let $f(x) = 2x^3 + x^2 + 2x + 2$ in $\mathbb{Z}_5[x]$. Then $f(0) = 2, f(1) = 2, f(-1) = -1, f(2) = 1$, and $f(-2) = 1$, so $f(x)$ has no zeros in $\mathbb{Z}_5$. Because $f(x)$ is of degree 3, Theorem 23.10 shows that $f(x)$ is irreducible over $\mathbb{Z}_5$.

14. $f(x) = x^2 + 8x - 2$ satisfies the Eisenstein condition for irreducibility over $\mathbb{Q}$ with $p = 2$. It is not irreducible over $\mathbb{R}$ because the quadriatic formula shows that is has the real zeros $(-8 \pm \sqrt{72})/2$. Of course it is not irreducible over $\mathbb{C}$ also.

15. The polynomial $g(x) = x^2 + 6x + 12$ is irreducible over $\mathbb{Q}$ because it satisfies the Eisenstein condition with $p = 3$. It is also irreducible over $\mathbb{R}$ because the quadratic formula shows that its zeros are $(-6 \pm \sqrt{-12})/2$, which are not in $\mathbb{R}$. It is not irreducible over $\mathbb{C}$, because its zeros lie in $\mathbb{C}$.

16. If $x^3 + 3x^2 - 8$ is reducible over $\mathbb{Q}$, then by Theorem 23.11, it factors in $\mathbb{Z}[x]$, and must therefore have a linear factor of the form $x - a$ in $\mathbb{Z}[x]$. Then a must be a zero of the polynomial and must divide -8, so the possibilities are $a = \pm 1, \pm 2, \pm 4, \pm 8$. Computing the polyomial at these eight values, we find none of them is a zero of the polynomial, which is therefore irreducible over $\mathbb{Q}$.

17. If $x^4 - 22x^2 + 1$ is reducible over $\mathbb{Z}$, then by Theorem 23.11, it factors in $\mathbb{Z}[x]$, and must therefore either have a linear factor in $\mathbb{Z}[x]$ or factor into two quadratics in $\mathbb{Z}[x]$. The only possibilites for a linear factor are $x \pm 1$, and clearly neither 1 nor -1 is a zero of the polynomial, so a linear factor is impossible. Suppose

$$x^4 - 22x^2 + 1 = (x^2 + ax + b)(x^2 + cx + d).$$

Equating coefficients, we see that

$x^3\ coefficient : 0 = a + c$

$x^2\ coefficient : -22 = ac + b + d$

$x\ coefficient: 0 = bc + ad$

$constant\ term: 1 = bd$ so either $b = d = 1$ or $b = d = -1$.

Suppose $b = d = 1$. Then $-22 = ac + 1 + 1$ so $ac = -24$. Because $a + c = 0$, we have $a = -c$, so $-c^2 = -24$ which is impossible for an integer c. Similarly, if $b = d = -1$, we deduce that $-c^2 = -20$, which is also impossible. Thus the polynomial is irreducible.

18. Yes, with $p = 3$. **19.** Yes, with $p = 3$.

20. No, for 2 divides the coefficient 4 of x^{10} and 3^2 divides the constant term -18.

21. Yes, with $p = 5$.

22. Let this polynomial be $f(x)$. If $f(x)$ has a rational zero, then this zero can be expressed as a fraction with numerator dividing 10 and denominator dividing 6. The possibilities are $\pm 10, \pm 5, \pm 10/3, \pm 5/2, \pm 2, \pm 5/3, \pm 1, \pm 5/6, \pm 2/3, \pm 1/2, \pm 1/3$, and $\pm 1/6$.

Experimentation with a calculator shows that there is a negative real zero between -2 and -3 because $f(-2) < 0$ and $f(-3) > 0$. (Recall the intermediate value theorem.) The only possible rational candidate is -5/2. We reach for our calculator and find that $f(-2.5) = 0$, so -5/2 is a zero and $(2x + 5)$ is a linear factor.

Because $f(0) < 0$ and $f(1) > 0$, the intermediate value theorem shows that there is a real zero a satisfying $0 < a < 1$. The rational possibilies are 5/6, 2/3, 1/2, 1/3, and 1/6. Because $2x + 5$ is a factor, accounting for the factor 2 of 6 and the factor 5 of 10, we can discard 5/6, 1/2, and 1/6, leaving 2/3 and 1/3 to try. We reach for our calculator and compute $f(2/3) = 0$, so $3x - 2$ is also a factor. Because we have accounted for the 6 and the 10 with these linear factors, the only other possible rational zeros would have to be 1 or -1, and we easily find that these are not zeros. Thus the rational zeros are 2/3 and -5/2.

23. The definition is incorrect. We must require that $g(x)$ and $h(x)$ have degree less than the degree of $f(x)$, and that the polynomial is nonconstant.

A nonconstant polynomial $f(x) \in F[x]$ is **irreducible over the field** F if $f(x) \neq g(x)h(x)$ for any polynomials $g(x), h(x) \in F[x]$ both of degree less than the degree of $f(x)$.

24. The definition is correct.

25. T T T F T F T T T T

26. Considering $f(x) = x^4 + x^3 + x^2 - x + 1$ in $\mathbb{Z}[x]$, we find that $f(-2) = 16 - 8 + 4 + 2 + 1 = 15$. Thus $p = 3$ and $p = 5$ are primes such that -2 is a zero of $f(x)$ in $\mathbb{Z}_p$, that is, such that $x + 2$ is a factor of $f(x)$ in $\mathbb{Z}_p[x]$.

27. The polynomials of degree 2 in $\mathbb{Z}_2[x]$ are

x^2: not irreducible because 0 is a zero,
$x^2 + 1$: not irreducible because 1 is a zero,
$x^2 + x$: not irreducible because 0 is a zero,
$x^2 + x + 1$: irreducible because neither 0 nor 1 are zeros.

Thus our answer is $x^2 + x + 1$.

28. The Polynomials of degree 3 in $\mathbb{Z}_2[x]$ are

x^3: not irreducible because 0 is a zero,
x^3+1: not irreducible because 1 is a zero,
x^3+x: not irreducible because 0 is a zero,
x^3+x^2: not irreducible because 0 is a zero,
x^3+x+1: irreducible, neither 0 nor 1 is a zero,
x^3+x^2+1: irreducible, neither 0 nor 1 is a zero,
x^3+x^2+x: not irreducible because 0 is a zero,
x^3+x^2+x+1: not irreducible, 1 is a zero.

Thus the irreducible cubics are x^3+x+1 and x^3+x^2+1.

29. The 18 polynomials of degree 2 in $\mathbb{Z}_3$ are

$x^2, x^2+x, x^2+2x, 2x^2, 2x^2+x, 2x^2+2x$ all reducible because 0 is a zero,

$x^2+2, x^2+x+1, 2x^2+1, 2x^2+2x+2$ all reducible because 1 is a zero,

$x^2+2x+1, 2x^2+x+2$ both reducible because 2 is a zero, and

$x^2+1, x^2+x+2, x^2+2x+2, 2x^2+2, 2x^2+x+1, 2x^2+2x+1$ which are all irreducible because none has 0, 1, or 2 as a zero.

30. An irreducible polynomial must have a nonzero constant term or 0 is a zero; this eliminates 18 of the 54 cubic polynomials in $\mathbb{Z}_3[x]$. Now a is a zero of $f(x)$ if and only if a is a zero of $2f(x)$, so we can consider just the 18 cubics with leading coefficient 1 and constant term nonzero.

$x^3+2,\ x^3+x^2+1,\ x^3+x+1,\ x^3+2x^2+2x+1,\ x^3+x^2+2x+2,\ x^3+2x^2+x+2$ have 1 as a zero, so they are reducible.

$x^3+1,\ x^3+2x^2+2,\ x^3+x+2,\ x^3+x^2+x+1$ have -1 as a zero, so they are reducible.

The remaining eight cubics with leading coefficient 1 and nonzero constant term, namely:

$x^3+2x+1,\ \ x^3+2x+2,\ \ x^3+x^2+2,\ \ x^3+2x^2+1,\ \ x^3+x^2+x+2,$
$x^3+x^2+2x+1,\ \ x^3+2x^2+x+1,\ \ \text{and}\ \ x^3+2x^2+2x+2$

and their doubles

$2x^3+x+2,\ \ 2x^3+x+1,\ \ 2x^3+2x^2+1,\ \ 2x^3+x^2+2,\ \ 2x^3+2x^2+2x+1,$
$2x^3+2x^2+x+2,\ \ 2x^3+x^2+2x+2,\ \ \text{and}\ \ 2x^3+x^2+x+1$

are irreducible.

31. Following the hint, each reducible quadratic that is of the form x^2+ax+b is a product $(x+c)(x+d)$ for $c, d \in \mathbb{Z}_p$. There are $\binom{p}{2} = p(p-1)/2$ such products (neglecting order of factors) where $c \neq d$. There are p such products where $c = d$. Thus there are $p(p-1)/2 + p = p^2/2 + p/2 = p(p+1)/2$ reducible quadratics with leading coefficient 1. Because the leading coefficient (upon multiplication) can be any one of $p-1$ nonzero elements, there are $(p-1)p(p+1)/2$ reducible quadratics altogether. The total number of quadratic polynomials in $\mathbb{Z}_p[x]$ is $(p-1)p^2$. Thus the number of irreducible quadratics is $(p-1)p^2 - (p-1)p(p+1)/2 = p(p-1)[p-(p+1)/2] = p(p-1)^2/2$.

32. Each zero of a polynomial leads to a linear factor, and the number of linear factors in the factorization of a polynomial cannot exceed the degree of the polynomial.

33. If the group were not cyclic, then the Fundamental Theorem for finitely generated abelian groups shows that the least common multiple m of the orders of the elements would be less than the number n of elements, leading to a polynomial $x^m - 1$ with $n > m$ zeros, which is impossible in a field.

34. Note that $x^2 = xx$ and $x^2 + 1 = (x + 1)^2$ are reducible in $\mathbb{Z}_p$. For an odd prime p and $a \in \mathbb{Z}_p$, we know that $(-a)^p + a = -a^p + a = -a + a = 0$ by Corollary 20.2. Thus $x^p + a$ has $-a$ as a zero, so it is reducible over $\mathbb{Z}_p$ for every prime p. [Actually, the binomial theorem and Corollary 20.2 show that $x^p + a = x^p + a^p = (x + a)^p$.]

35. We are given that $f(a) = a_0 + a_1a + \cdots + a_na^n = 0$ and $a \neq 0$. Dividing by a^n, we find that

$$a_0(\frac{1}{a})^n + a_1(\frac{1}{a})^{n-1} + \cdots + a_n = 0$$

which is just what we wanted to show.

36. By Theorem 23.1, we know that $f(x) = q(x)(x - a) + c$ for some constant $c \in F$. Applying the evaluation homomorphism ϕ_a to both sides of this equation, we find that

$$f(a) = q(a)(a - a) + c = q(a)0 + c = c,$$

so the remainder $r(x) = c$ is actually $f(a)$.

37. **a.** Let $f(x) = \sum_{i=0}^{\infty} a_ix^i$ and $g(x) = \sum_{i=0}^{\infty} b_ix^i$. Then

$$\begin{aligned}\overline{\sigma_m}(f(x) + g(x)) &= \overline{\sigma_m}\left(\sum_{i=0}^{\infty}(a_i + b_i)x^i\right) = \sum_{i=0}^{\infty}\overline{\sigma_m}\,(a_i + b_i)x^i \\ &= \sum_{i=0}^{\infty}[\overline{\sigma_m}\,(a_i) + \overline{\sigma_m}\,(b_i)]x^i = \overline{\sigma_m}\,(f(x)) + \overline{\sigma_m}\,(g(x))\end{aligned}$$

and

$$\begin{aligned}\overline{\sigma_m}\,(f(x)g(x)) &= \overline{\sigma_m}\left(\sum_{n=0}^{\infty}\left(\sum_{i=0}^{n}a_ib_{n-i}\right)x^n\right) = \sum_{n=0}^{\infty}\overline{\sigma_m}\left(\sum_{i=0}^{n}a_ib_{n-i}\right)x^n \\ &= \sum_{n=0}^{\infty}\left(\sum_{i=0}^{n}\overline{\sigma_m}\,(a_ib_{n-i})\right)x^n = \sum_{n=0}^{\infty}\left(\sum_{i=0}^{n}\overline{\sigma_m}\,(a_i)\overline{\sigma_m}\,(b_{n-i})\right)x^n \\ &= \overline{\sigma_m}\,(f(x))\overline{\sigma_m}\,(g(x)),\end{aligned}$$

so $\overline{\sigma_m}$ is a homomorphism. If $h(x) \in \mathbb{Z}_m[x]$, then if $k(x)$ is the polynomial in $\mathbb{Z}[x]$ obtained from $h(x)$ by just viewing the coefficients as elements of $\mathbb{Z}$ rather than of $\mathbb{Z}_m$, we see that $\overline{\sigma_m}\,(k(x)) = h(x)$, so the homomorphism $\overline{\sigma_m}$ is onto $\mathbb{Z}_m[x]$.

b. Let $f(x) = g(x)h(x)$ for $g(x), h(x) \in \mathbb{Z}[x]$ with the degrees of both $g(x)$ and $h(x)$ less than the degree n of $f(x)$. Applying the homomorphism $\overline{\sigma_m}$, we see that $\overline{\sigma_m}\,(f(x)) = \overline{\sigma_m}\,(g(x))\overline{\sigma_m}\,(h(x))$ is a factorization of $\overline{\sigma_m}\,(f(x))$ into two polynomials of degree less than the degree n of $\overline{\sigma_m}\,(f(x))$, contrary to hypothesis. Thus $f(x)$ is irreducible in $\mathbb{Z}[x]$, and hence in $\mathbb{Q}[x]$ by Theorem 23.11.

c. Taking $m = 5$, we see that $\overline{\sigma_5}\,(x^3 + 17x + 36) = x^3 + 2x + 1$ which does not have any of the five elements 0, 1, -1, 2, -2 of $\mathbb{Z}_5$ as a zero, and is thus irreducible over $\mathbb{Z}_5$ by Theorem 23.10. By Part(**b**), we conclude that $x^3 + 17x + 36$ is irreducible over $\mathbb{Q}$.